NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment
Introduction

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About TET4185:Power Markets, Resources and Environment

- **Course Focus:** Explores electricity market principles with an emphasis on Norwegian/Nordic solutions, integrating technical aspects of power systems, optimization, and economics, underpinned by mathematical rigor.
- **Learning Outcomes:**
  - **Knowledge:** Understanding deregulated power markets, market behavior, and price dynamics; analyzing electricity networks and their impact on market clearing; evaluating carbon reduction tools.
  - **Skills:** Formulating market simulation problems, managing transmission congestion, and performing optimization-based dispatch and power flow calculations.
- **Methodology:** Combines lectures, exercises, and project assignments, emphasizing practical application through assignments.

Teachers

- **Course coordinator :**
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- **Research Assistant:**
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- **Student assistants**
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  - Heidi Elisabeth Havn, heidi.e.havn@ntnu.no
  - Stina Hinnerichsen, stina.hinnerichsen@ntnu.no
- **Co-lecturers**
  - Kasper Emil Thorvaldsen (SINTEF Energy and IEL)
  - Pieter Schavemaker (E-Bridge)

SINTEF

E-Bridge
Kompetenz in Energie

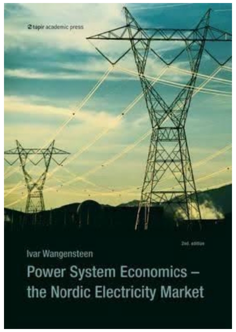
Background knowledge

- TET4105 Power system analysis 1
- TET4135 Energy system planning and operation
- Some background in operations research like TIØ4120 Operations Research, is recommended.

- Note: Despite courses being recommended, some background information on the topics will be given in class

Curriculum

- The main curriculum is: **Power System Economics: The Nordic Electricity Market**, Ivar Wangensteen, Second Edition, 2012
- Ch. 3 & 4, **Fundamentals of Power System Economics**, Daniel S. Kirschen and Goran Strbac.
- The lectures are sometimes significantly extended compared with the book.

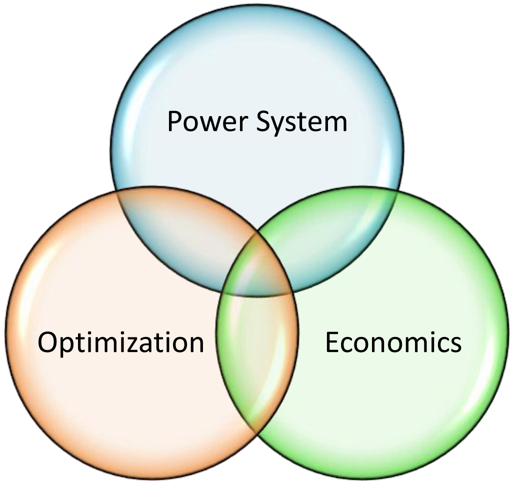


Everything that is in the lecture slides is also part of the curriculum, unless it is explicitly stated that it is not!

An Interdisciplinary Approach

Three Pillars of Power Market Studies

- **Power System**
  - Description: Focus on the technical aspects of electricity generation, transmission, and distribution.
  - Key Concepts: Power flow, grid stability, renewable integration.
- **Economics**
  - Description: Explore the financial and regulatory frameworks that underpin the energy markets.
  - Key Concepts: Market structures, pricing strategies, economic incentives.
- **Optimization**
  - Description: Application of mathematical methods to achieve optimal decisions and efficiency.
  - Key Concepts: Cost minimization, resource allocation, optimization models.



The overview of the course:

- Basic of microeconomics
- Market for Electrical Energy Participating in the Market
- Pricing & Tariff structure
- Restructuring and Nordic Market
- Grid access, Congestion management
- Transmission pricing based on OPF I and II
- Flow-Based Market Coupling
- Ancillary services
- Environmental policy
- Hydro Generation

Lecture Plan, Tuesday 11:15 – 14:00 (VE1 Verkstedteknisk)

Week no	Lecture #	Date	Lecture title	Chapter and Lecturer
3	1	**January 13**	Introduction Basic microeconomics	Ch1 & 2, Farahmand
4	2	**January 20**	Market for Electrical Energy Participating in the Market	Ch. 3 & 4: Power System Economics, Farahmand
5	3	**January 27**	Introduction to LP problem (1h) Introduction to Python/Pyomo (2h)	Lecture slides and note, Farahmand Lecture slides and note, Thorvaldsen
6	4	**February 03**	THEMA (1h) Pricing (2h)	Ch4, Farahmand
7	5	**February 10**	Restructuring and Nordic Market Optimum mix of generating units	Ch 5, Farahmand Ch 6, Farahmand
8	6	**February 17**	Grid access, Congestion management	Ch 7, Farahmand
9	7	**February 24**	Transmission pricing based on OPF(1)	Ch7, Farahmand
10	8	**March 03**	Transmission pricing based on OPF(2)	Ch7 + Lecture notes, Farahmand
11	9	**March 10**	Flow-Based Market Coupling	Lecture notes, Pieter Schavemaker, E-Bridge
12	10	**March 17**	Environmental policy	Ch 10, Farahmand
13	11	**March 24**	Ancillary services	Ch 8 with updates, Farahmand
15	12	**April 07**	Hydro Generation	Ch 6, Farahmand Farahmand
16	13	**April 14**	Summing up, questions	Farahmand

Assignments, Thursdays 10:15 – 12:00 (GL-GEL EL3)

Week no	Date	Assignment	Deadline
4	**January 22**	Assignment 1. Perfect & Imperfect competition	Jan 29
5	**January 29**	Assignment 2. Use of Nord Pool's markets	Feb 5
6	**February 05**	Assignment 3. Power generation	Feb 12
7	**February 12**	Assignment 4, Optimum mix of generating capacity	Feb19
8	**February 19**	Assignment 5. Calculation of system and area prices	Feb 26
9	**February 26**	Assignment 6. Transmission pricing based on OPF	Mar 02
10	**March 02**	**Project assignment**	**May 4**
11	**March 12**	Assignment 7. Flow-Based market coupling	Mar 19
12	**March 19**	Assignment 8. Environmental Policy	Mar 26
13	**March 26**	**Break**	
14	**April 09**	Assignment 9. Hydropower modelling	Apr 14
19	Oral presentations of the projects		May 7-8

9 assignments and at least 6 assignments have to be approved. Students will have to get at least one assignment approved among assignments 7 to 9.

The course evaluation

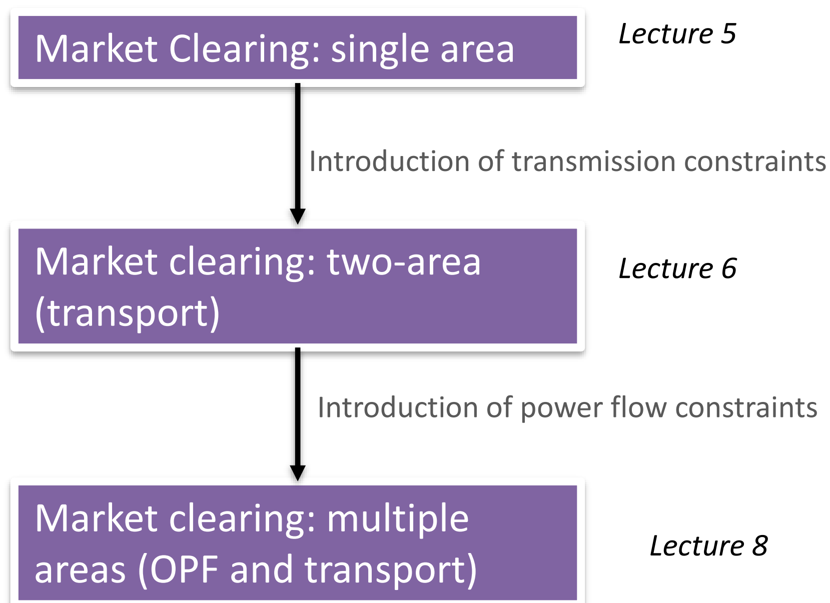
- The final evaluation will be based on the **group project assignment (3 persons)**. Final grading will be from A to F.
  - Project will be handed out on March 10th, deadline May 5th.
  - If failed in the project assignment, the student should resubmit an improved project in a re-sit submission.
- Students must pass the exercise criteria for being admissible for evaluation
- <u>80% </u>written project, <u>20% </u>oral presentation of project

Programming skills

- **Python** will be introduced as a programming language
  - Commonly used within this field
  - Very similar to MATLAB
- **Introduction course given on January 27th**
- Exercises will include programming tasks later
- We will also use the optimization model tool PYOMO for Python to solve scheduling problems and market clearing (all code will be given)
- Python/Pyomo will be relevant and necessary for the course project

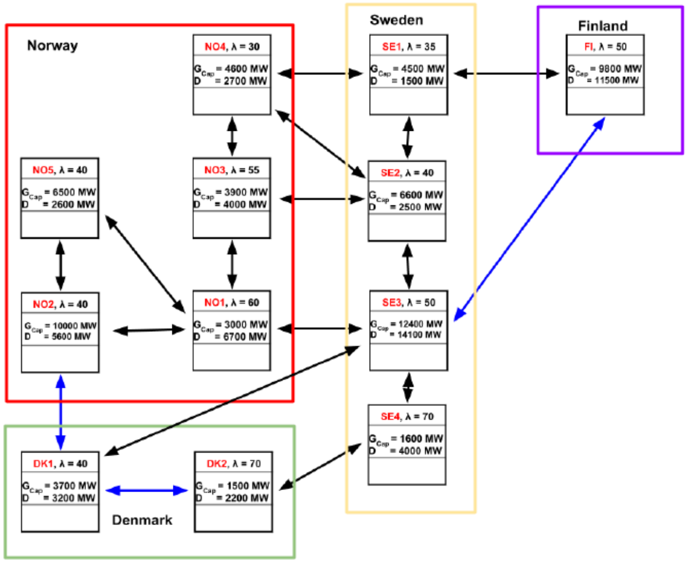
Modelling market clearing as an optimization problem

- **Objective**: Model and solve market clearing in power systems using optimization techniques.
- **Approach**: Emphasis on optimization coding to develop practical skills in formulating and solving complex market models.
- Students will be able to model various market clearing scenarios as optimization problems, implement these models in code, and critically analyze the outcomes to make informed decisions in the context of power economics.



Project assignment (100 % of total grade)

- Will be on Optimal Power Flow (OPF) using DCOPF approximation.
- Workload will be:
  - Part 1: Understanding DCOPF and environmental policy
  - Part 2: Modelling the Nordic System with DCOPF (code given)
  - Part 3: Extend the model with a specific task
  - Part 4: Environmental policy
- The modelling will be in Python/Pyomo
- Groups of <u>3 persons</u>
- <u>80%</u> written project, <u>20%</u> oral presentation
- Presentation of the project (5 min)
- More detail on this will be given later in the semester



Reference group for the course

- The course is in need of students willing to participate in the reference group
- Ensures that the course has good quality and represents the class as a whole. 1-2 meetings per semester
- Need candidates from each study program
  – MTENERG, MTIØT, International Masters, others
- Receive a diploma at the end, good material on your CV

The reference group's tasks

The reference group should have an **ongoing dialogue** with other students throughout the semester.

- The reference group may also choose to take other measures such as hold student meetings or conduct surveys before reference group meetings.

The reference group should **represent all students** enrolled in the course at the reference group meetings.

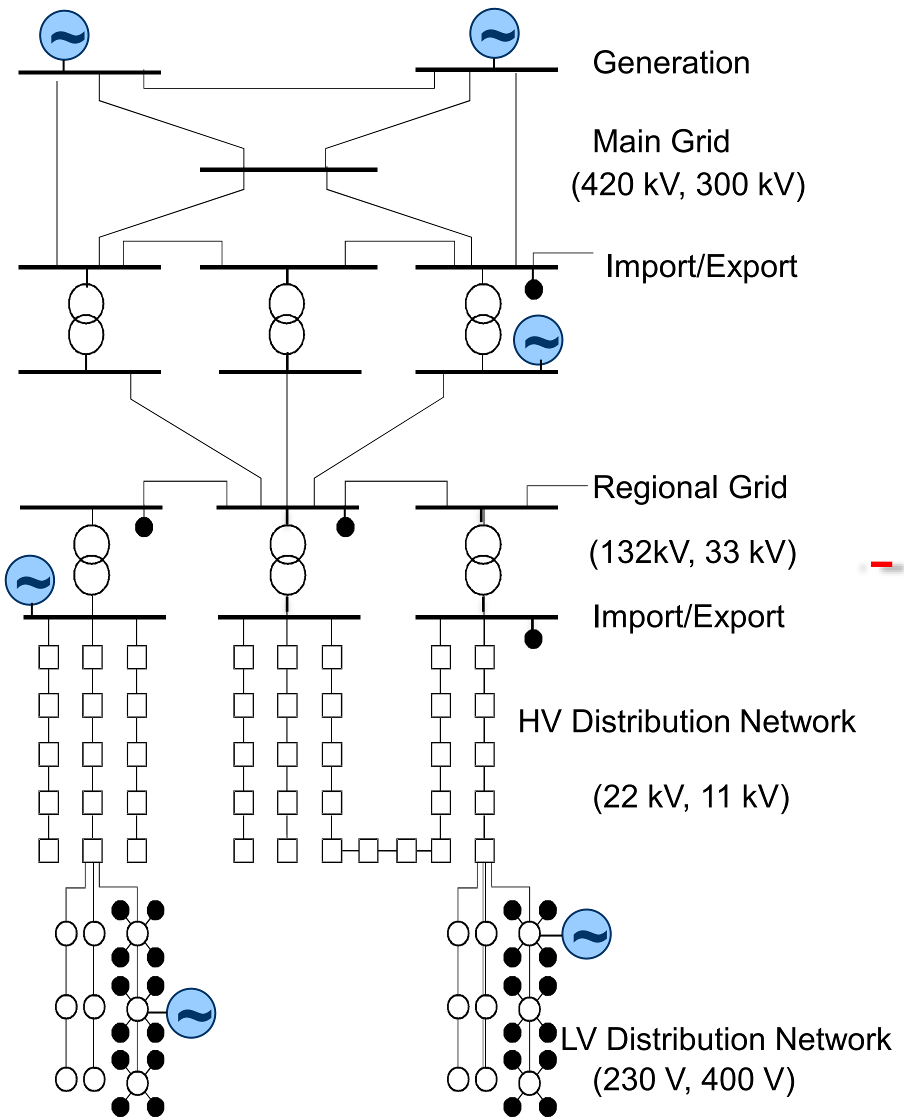
Introduction

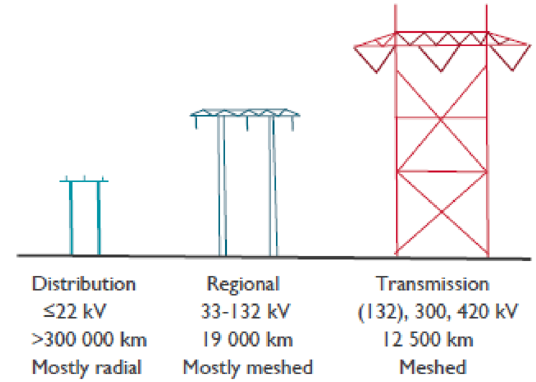
Perspective (some warmup)

- Physical system and organization of the power sector
- Restructuring – the reasons behind today's system
- Reasons for restructuring
- Electricity is special
- Market design
- Criticism of restructuring

- **Microeconomics (Supply, demand, demand elasticity)**
- **Different forms of competition**

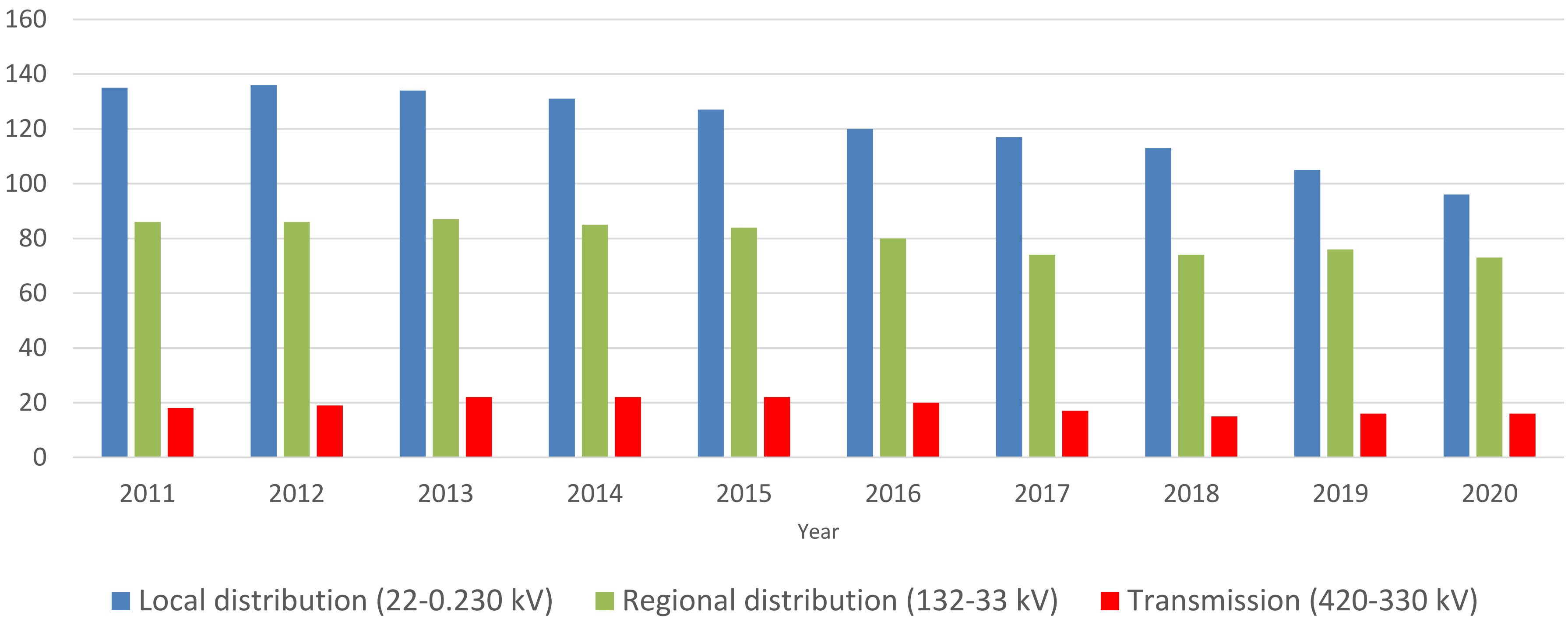
Physical system





Network Code	Norway
Transmission Grid (TSO)	Sentralnett
	~~Regionalnett~~
Distribution Grid (DSO)	Distribusjonsnett

Grid levels and companies (2020)

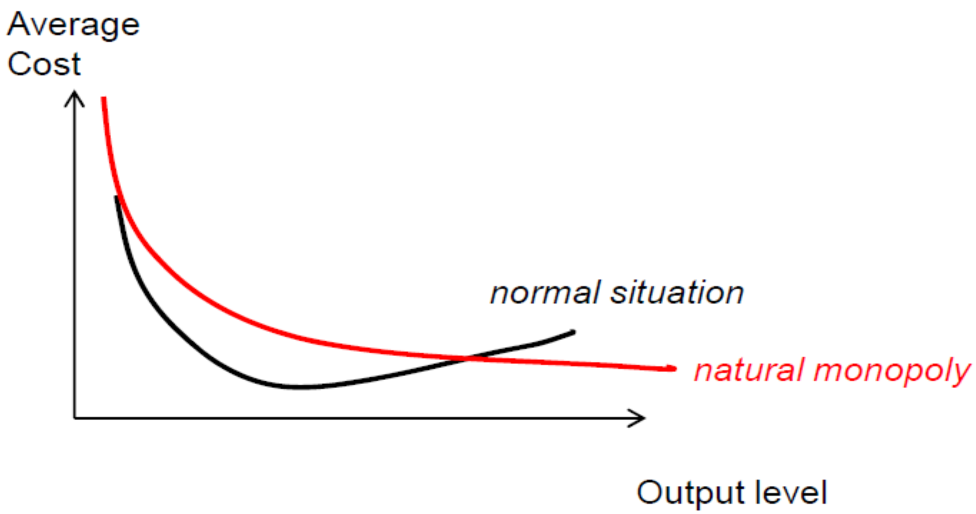


NVE, The Norwegian Water Resources and Energy Directorate (https://www.nve.no)

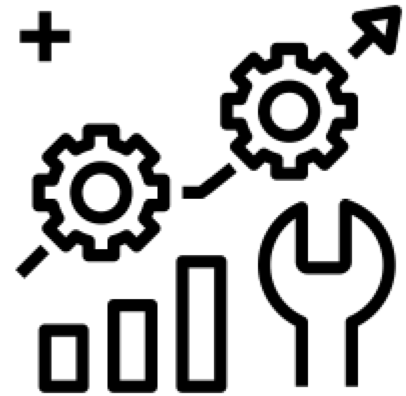
Historical organization of power sector

Natural **monopoly** (one large company that owned the whole power sector)

- ***High Fixed Costs***: The initial investment required to build the infrastructure (e.g., power grids, water supply systems, railways) is substantial, making duplication by competitors inefficient.
- **Economies of Scale**: As production increases, the average cost per unit decreases, favoring a single provider
- **Essential Services**: Often involves critical public utilities like electricity, water, natural gas, or public transportation.



Restructuring



- Traditional organization:
  - Privately owned with public regulation (US)
  - Publicly owned (Europe)
    - One large company (UK, France, Italy,…)
    - Many public companies (Norway, Netherlands…)

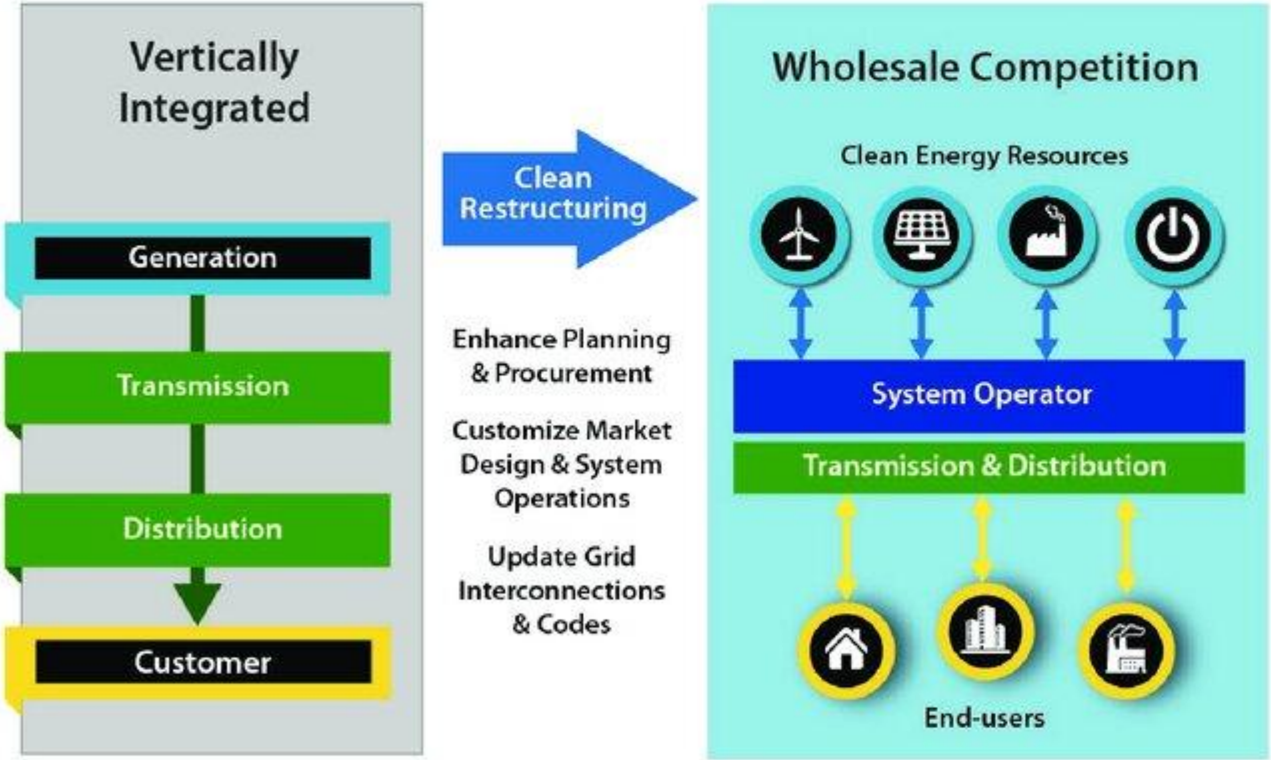
- Deregulation / restructuring / liberalization /privatization
  - UK (1980), Norway (Energy Act of 1990)[1]
- Unbundling
  - Grids have natural monopoly characteristics
  - But production and retail have not

[1] https://www.regjeringen.no/en/topics/energy/electricity/the-power-market-and-prices/id2076000/

Restructuring

Introduction of Competition in Generation and Retail:

- Restructuring aims to move away from vertically integrated monopolies by separating electricity generation, transmission/distribution and retail.
- By introducing competition in generation (wholesale markets) and retail (consumer choice), the objective is to improve efficiency, reduce costs, and foster innovation.



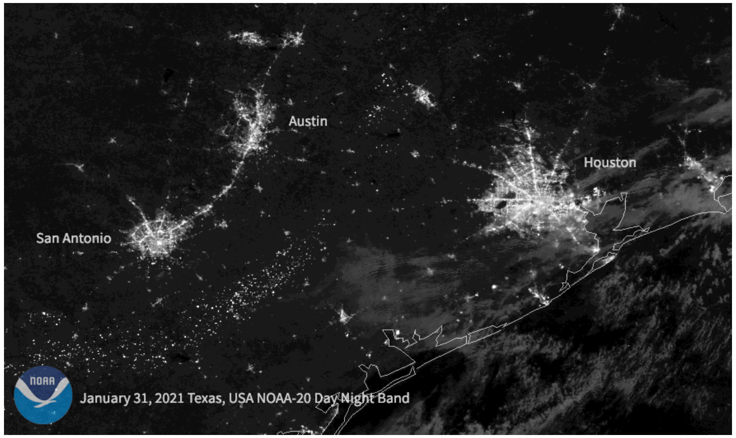
Why "restructure"? =competition in generation and retail

Goal is to provide long run net benefits to society

- **Efficiency Gains**: Better incentives for controlling capital and operating costs of new and existing generating capacity
- **Encourage innovation** in power supply technologies
- Shift risks of “mistakes” to suppliers and away from consumers
- **Price Signals**: Support retail prices that reflect marginal production cost including the costs of congestion, losses, and scarcity
- **Consumer Choice**: Provide retail service products based on individual consumer preferences
- **Facilitate better regulation** of residual monopoly to enhance efficiency in T&D
- **Environmental and Policy Goals**: Consistent with environmental and reliability policy goals using market-compatible mechanisms (green taxes or cap-and-trade)

Special features of electricity

- ***Momentary balance between generation and consumption***
- ***Non-storability***
- ***Consumption variability*** *(typical daily, weekly, annual patterns)*
- ***Inelasticity of consumption***
- ***Non-traceability***
- ***Essential to the community***
- ***Breakdown possibility***



2021 Texas power crisis- At least 210 people lost their lives during the winter storm

27 September 2006

57 mill. Italians without power for several hours

The April 2025 Iberian blackout started on 28 April 2025 around 12:30 local time, lasted about 10–12 hours with restoration by early 29 April, and affected around 57–58 million people across Spain and Portugal.

Features of good market design[1]
short-term operation

- **Efficient Scheduling**: Balance generation and load continuously to match supply with demand (security of supply) while minimizing costs.
- **Optimal Use of Transmission Capacity:** Allocate scarce transmission capacity efficiently to maximize the security of power supply.
- **Management of Operating Reserves:** Effectively manage reserves to address emergencies and unplanned outages in generation and transmission through market mechanisms.
- **System Coordination**: Coordinate with neighboring systems to enhance trade and system reliability.

[1]Market design refers to a set of rules that determine how the market is run

Features of good market design
Long-term Investment Incentives

Well-functioning wholesale markets are necessary to provide appropriate:

- Incentives for investment in the new **generation** using several revenue streams
  - energy revenues
  - ancillary services or “operating reserves” revenues
  - capacity revenues (if any)

- Signals for **transmission** investments using
  - congestion management and locational pricing
  - contract regime (spot, longer term)
  - transmission interconnection and transit rules and prices

Many markets

- Wholesale
  - Producers, large consumers, retailers
- Retail
  - Small consumers, prosumers

- Financial days → years
- Day-ahead next day
- Intraday same day (coming hour)
- Balancing Now!

- Different products
  - Capacity next day → year
  - Energy real-time

Criticism of deregulation

- **High prices**
  - In some periods prices tend to be higher than in regulated markets.
  - Perceived as unfair and unequal among consumers
- **Volatile prices:** Greater fluctuations in prices can occur due to market dynamics
- **Market power:** Few dominant players can manipulate market prices and conditions.
- **Inefficient and costly regulation:** Regulations intended to control the residual monopoly can become complex and expensive.
- **Reliability concerns:** Potential risks to grid stability and electricity supply reliability.
- **Environmental concerns:** Deregulated markets might not adequately address environmental concerns, leading to higher emissions.
- **Lacking investment incentives for generation:** Lack of clear incentives for new generation capacity, which could hinder long-term sustainability.